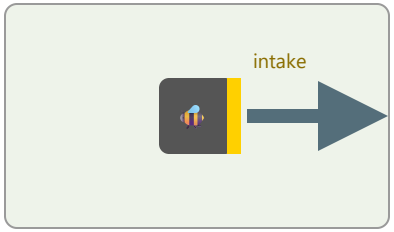
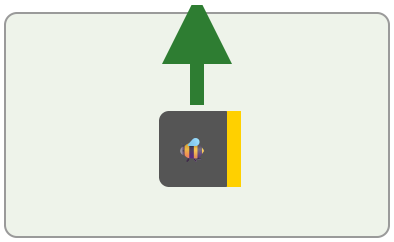
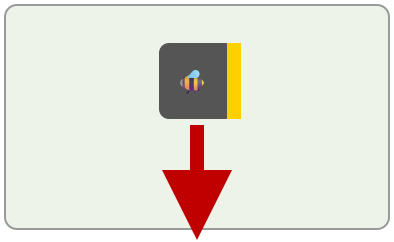
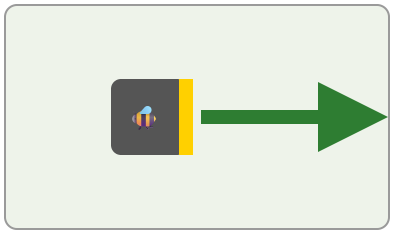
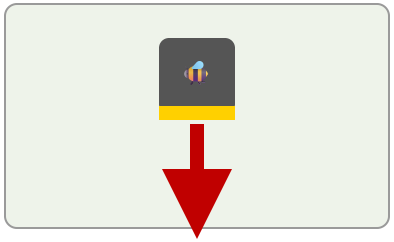
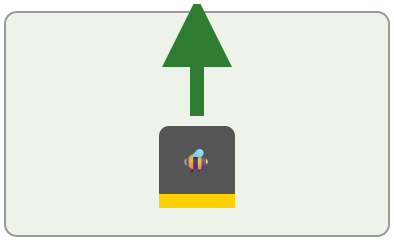
FIELD-CENTRIC DRIVING: what changes

Team 37102 · BioBuzz TeleOp · read with the Driver Card

Robot-centric is the default when the robot app starts: the stick works from the **robot's** point of view. **FIELD-CENTRIC** (optional) makes the stick work from **YOUR** point of view, whichever way the robot faces. **Check the INIT screen's Drive line every match**: the choice is kept until the robot app restarts.

	ROBOT-CENTRIC (default)	FIELD-CENTRIC (optional)
↑ Left stick UP robot faces right	 **YOU (drivers) ▼ here** toward its **intake** (to the right here)	 **YOU (drivers) ▼ here** **away from you ↑**, whichever way it faces
→ Left stick RIGHT robot faces right	 **YOU (drivers) ▼ here** the **robot's** right = **toward you ↓**	 **YOU (drivers) ▼ here** **your** right →
↑ Left stick UP intake faces YOU	 **YOU (drivers) ▼ here** toward its intake = **toward you ↓**	 **YOU (drivers) ▼ here** **still away from you ↑**

	Robot-centric	Field-centric
Left stick ↑	toward the **intake**	**away from you**
Left stick →	the **robot's** right	**your** right
Right stick	turns the robot: **the same**	
Hold L1	precision (slow) drive: **the same**	
Aiming	While you **hold ✕ or R1** and our CELL is in view, the stick is **robot-centric in both** (sideways = sideways to the CELL). Let go and field-centric is back at once. (If the CELL drops out of view while you still hold, it waits 0.5 s.)	
Every other button	unchanged (Driver Card)	

The key idea

With the **intake pointing away from you**, both modes drive **exactly the same**. The difference shows when the robot is **turned**: sideways, or intake toward you. Then robot-centric makes you think "where is the intake?", and field-centric doesn't.

When is it worth it? Coming back from the GARDEN or the LOADING ZONE with the robot turned, dodging a defender while spinning, lining up sideways. Pick **one mode per match** with the coach, and only use field-centric after the practice drills (page 2).

"Away from you" = straight across the field from where **you** stand at the alliance station, not "toward the HIVE" and not "the robot's front".

FIELD-CENTRIC: how to use it

turn on · set "away" · recover · turn off

1 🎮 Turn it ON in INIT

Read the `Drive` line. **L1 switches** it (it's not "on" every press): press until it says `FIELD-CENTRIC`.

2 🧭 "Away" is set

After our AUTO (less than 60 s before): automatic, the screen says `Start heading: from AUTO`. Don't press OPTIONS.
Practice: start with the **intake pointing away** from you.

3 ✋ Hands off in INIT

The gyro calibrates. Screen: `Pinpoint gyro: ok`

4 ⬆ Drive

Stick ⬆ = **away from you**, whichever way the robot faces. Turn with the right stick as usual.

5 🔄 Feels wrong?

Point the **intake away** from you, then **TAP OPTIONS** (one buzz). Fixed.

TAP OPTIONS (in the match: let go before 1 s)

"The intake points **away from me NOW**": one buzz when you let go. Use it after a big hit, a hub restart, or whenever ⬆ doesn't go away from you. **Only tap with the intake really pointing away from you. In INIT** a press sets "away" at once (no buzz) and replaces AUTO's heading.

HOLD OPTIONS 1 s (in the match)

Field-centric **ON** = **1 blip** · **OFF** (robot-centric) = **2 blips**. It doesn't change "away". (In INIT, use L1 instead.) **Panic button:** if field-centric ever confuses you, hold OPTIONS 1 s and drive robot-centric.

The Driver Hub screen

`robot-centric (hold Options 1 s: field-centric)`
off (the default)

`FIELD-CENTRIC, intake +35 deg from away (PINPOINT)`
on and working (the robot is turned 35° left of "away")

`…, robot-centric while aiming`
✕ / R1 is aiming at the CELL: normal

`robot-centric: NO HEADING`
no gyro reading: it drives robot-centric by itself and goes back to field-centric when the reading returns. Only tap OPTIONS if ⬆ is wrong after that.

Don't

- **Move the robot in INIT.** The gyro then drifts all match and ⬆ slowly turns.
- **Press OPTIONS in INIT after AUTO**, unless the intake points away from you: it replaces AUTO's heading.
- **Hold** OPTIONS when you mean to tap: holding switches the mode (count the blips).
- **Forget to check INIT.** The mode stays from the last TeleOp; after a power cycle it's robot-centric.

Self-check

1. Field-centric, intake facing you: stick ⬆ goes… (away from you)
2. Tapped OPTIONS with the intake pointing left: ⬆ goes… (left; point it away, tap again)
3. Quick way back to robot-centric? (hold OPTIONS 1 s: 2 blips)
4. Our AUTO just ran: how do you set "away"? (nothing: it comes from AUTO)

Practice drills (10 min)

1. **Square:** robot sideways (intake right). Stick ⬆ ➡ ⬇ ⬅ without turning: the robot goes up, right, down, left on the field.
2. **Spin and go:** hold stick ⬆ and turn with the right stick. The robot keeps moving away from you while it spins.
3. **Break it, fix it:** intake pointing left, **tap** OPTIONS (wrong on purpose): ⬆ now goes left. Point the intake away, tap again: fixed.
4. **Panic button:** while driving, hold OPTIONS 1 s (2 blips: robot-centric), then again (1 blip: field-centric).
5. **Aim:** hold ✕ with the CELL in view and push the stick sideways: the robot slides sideways to the CELL.